Chapter 4: Grover’s Algorithm

This lesson is based on Grover's algorithm. It is a search algorithm for unstructured search problems. Quantum Algorithm Zoo

COntent:

1. Unstructured search
2. Grover’s Algorithm
   - Algorithm description
   - Analysis
3. Choosing the number of iterations
   - Unique search
   - Multiple Solutions
4. Conclusion
5. Implementation on Qiskit
   - Grover Algorithm
   - Precision depending of the number of solution

1. Unstructered search

We set $\Sigma = \{0, 1\}$ which corresponds to the binary alphabet.

We assume that we have a function f that does:

$$f : \Sigma^n \to \Sigma$$

which can be calculated efficiently.

Our goal is to find a solution that is a binary string $x \in \Sigma^n$ for $f(x) = 1$.

Search algorithm:

> Input: $f : \Sigma^n \to \Sigma$
> Output: a string $a \in \Sigma^n$ satisfying $f(x) = 1$, or no solution if no such string exists.

This is an unstructured problem because f is arbitrary. There is no promise, and we cannot count on it to obtain a structure that allows us to obtain solutions easily.

We note:

$$N = 2^n$$

When a letter is capitalized, it corresponds to the power 2 of the lowercase letter.

By iterating over all $x \in \Sigma$ and evaluating f for each one, we can solve the search

with N queries.

This is the best we can do with a deterministic algorithm.

Probabilistic algorithms offer minimal improvements, but require a number of queries that is linear to N. Grover's algorithm is a search algorithm that requires a number of queries in $O(\sqrt{N})$.

Phase query gates:

We assume that we have access to a function $f : \Sigma^n \to \Sigma$ through a query gate.

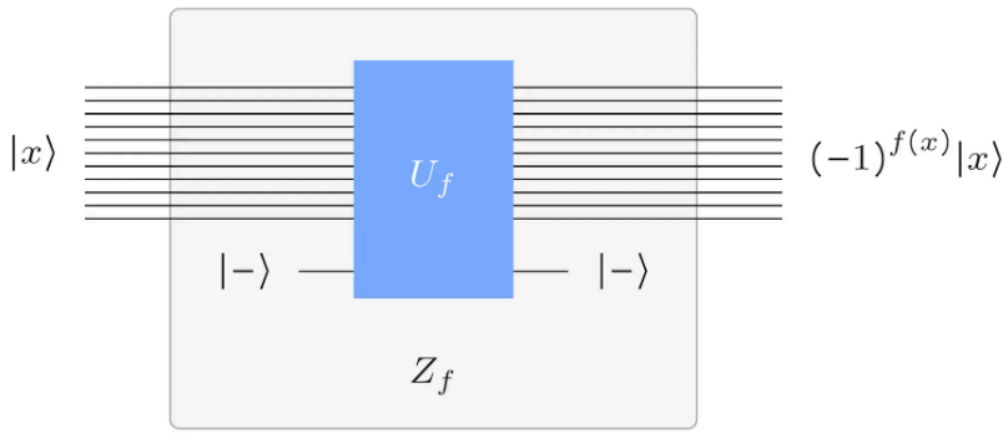
$$U_f : |a\rangle|x\rangle \to |a \oplus f(x)\rangle|x\rangle \;\; (\forall x \in \Sigma \text{ and } x \in \Sigma^n)$$

We can construct a circuit U_f for a Boolean circuit for f.

A “phase query gate” or phase query gate for f operates as follows:

$$Z_f : |x\rangle \to (-1)^{f(x)}|x\rangle \; (\forall x \in \Sigma^n)$$

Circuit example:



With a control qubit, it is possible to reconvert U_f, otherwise not. Need for a phase query gate for n-bit OR function:

$$(\forall x \in \Sigma^n) \;\; OR(x) = \begin{cases} 0 & x = 0^n \\ 1 & x \neq 0^n \end{cases} \tag{1}$$

$$(\forall x \in \Sigma^n) \;\; ZOR(x) = \begin{cases} |x> & x = 0^n \\ -|x> & x \neq 0^n \end{cases} \tag{2}$$

2. Grover’s Algorithm

Algorithm:

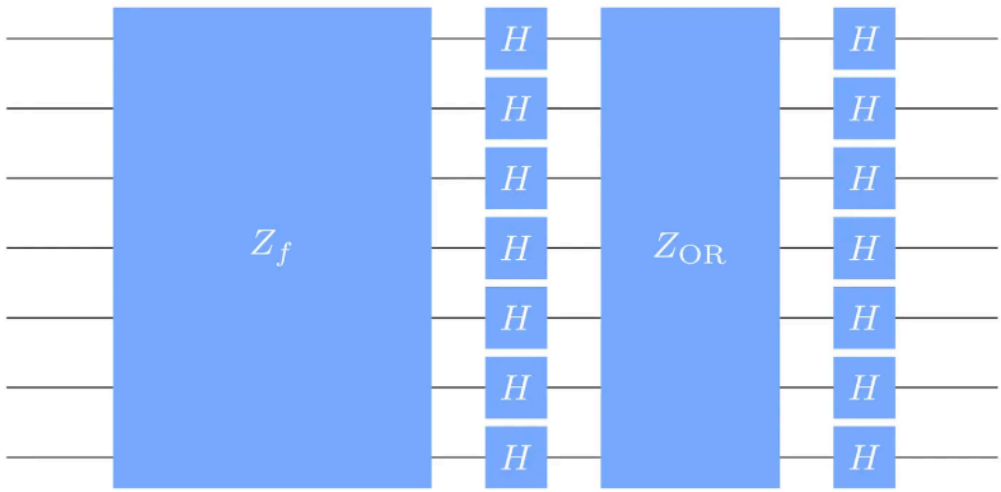
> 1. Initialize: Set n qubits in the state $H^{\otimes n}|0^n>$
> 2. Iteration: Apply Grover's operation t times (t is specialized later)

> 3. Measurement: A standard measurement basis provides a candidate solution.

Grover's operator is defined as follows:

$$G = H^{\otimes n} Z_{OR} H^{\otimes n} Z_f$$

Z_f is the query phase gate for f and Z_{OR} is the query phase gate for the n-bit OR function.



A typical Grover algorithm can be applied as follows:

> 1. Choose a number of iterations t
> 2. Run Grover's algorithm with t iterations to obtain a candidate solution x.
> 3. Check the solution. If $f(x) = 1$, then output x; otherwise, run Grover's algorithm again (if available with a different t); otherwise, there are no solutions.

Solutions and non-solutions:
We refer to Grover's algorithm for n qubits as a Q register.

We are interested in what happens when Q is initialized in the state $H^{\otimes n}|0^n\rangle$ and Grover's operator G is used iteratively.

$$G = H^{\otimes n} Z_{OR} H^{\otimes n} Z_f$$

The sets of non-solutions and solutions are:

> $$A_0 = \{x \in \Sigma^n : f(x) = 0\}$$
> $$A_1 = \{x \in \Sigma^n : f(x) = 1\}$$

A_0 contains all non-solutions and A_1 contains all solutions.

We may be interested in the set of superpositions of these sets:

> $$|A_0\rangle = \frac{1}{\sqrt{|A_0|}} \sum_{x \in A_0} |x\rangle$$
>
> $$|A_1\rangle = \frac{1}{\sqrt{|A_1|}} \sum_{x \in A_1} |x\rangle$$

Of course, we assume that these sets are not empty.

Analysis:

The Q register is initially initialized:

$$|u\rangle = H^{\otimes n}|0^n\rangle \ = \frac{1}{\sqrt{N}} \sum_{x \in \Sigma^n} |x\rangle$$

This state is contained in the extended subspace of $|A_0\rangle$ and $|A_1\rangle$:

$$|u\rangle = \sqrt{\frac{|A_0|}{N}}|A_0\rangle + \sqrt{\frac{|A_1|}{N}}|A_1\rangle$$

The state of Q remains in the subspace after each application of the Grover operator G.

Action of the Grover operation:

$$G = (H^{\otimes n} Z_{OR} H^{\otimes n})(Z_f)$$

1. Reminder:

> $$Z_f : |x\rangle \to (-1)^{f(x)}|x\rangle \ (\forall x \in \Sigma^n)$$
>
> The actions of Z_f on $|A_1\rangle$ and $|A_0\rangle$ are:
>
> $$Z_f|A_0\rangle = |A_0\rangle$$
>
> $$Z_f|A_1\rangle = -|A_1\rangle$$

2. The operation Z_{OR} is defined as follows:

> $$ZOR(x) = \begin{cases} |x\rangle & x = 0^n \\ -|x\rangle & x \neq 0^n \end{cases} \tag{3}$$
>
> There is an alternative notation for Z_f:
>
> $$Z_{OR} = 2|0^n\rangle\langle 0^n| - \mathbb{I}$$
>
> Using this expression, we can rewrite $H^{\otimes n} Z_{OR} H^{\otimes n}$ as:
>
> $$H^{\otimes n} Z_{OR} H^{\otimes n} = H^{\otimes n}(2|0^n\rangle\langle 0^n| - \mathbb{I})H^{\otimes n} = 2|u\rangle\langle u| - \mathbb{I}$$

Then:

$$G|A_0\rangle = (2|u\rangle\langle u| - \mathbb{I})Z_f|A_0\rangle$$
$$= (2|u\rangle\langle u| - \mathbb{I})|A_0\rangle$$
$$= 2\sqrt{\frac{|A_0|}{N}}|u\rangle - |A_0\rangle$$
$$= 2\sqrt{\frac{|A_0|}{N}}(\sqrt{\frac{|A_0|}{N}}|A_0\rangle + \sqrt{\frac{|A_1|}{N}}|A_1\rangle) - |A_0\rangle$$
$$= \frac{\sqrt{|A_0| - |A_1|}}{N}|A_0\rangle + 2\frac{\sqrt{|A_0|.|A_1|}}{N}|A_1\rangle$$

Finally:

$$G|A_0\rangle = \frac{|A_0| - |A_1|}{N}|A_0\rangle + 2\frac{\sqrt{|A_0|.|A_1|}}{N}|A_1\rangle$$

Similarly, for $G|A_1\rangle$, we obtain:

$$G|A_1\rangle = -2\frac{\sqrt{|A_0|.|A_1|}}{N}|A_0\rangle + \frac{|A_0| - |A_1|}{N}|A_1\rangle$$

The action of G on the set $\{|A_0\rangle, |A_1\rangle\}$ can be described as a square matrix of dimension 2:

$$M = \begin{pmatrix} \frac{|A_0|-|A_1|}{N} & -2\frac{\sqrt{|A_0|.|A_1|}}{N} \\ 2\frac{\sqrt{|A_0|.|A_1|}}{N} & \frac{|A_0|-|A_1|}{N} \end{pmatrix} = \begin{pmatrix} \sqrt{\frac{|A_0|}{N}} & -\sqrt{\frac{|A_1|}{N}} \\ \sqrt{\frac{|A_1|}{N}} & \sqrt{\frac{|A_0|}{N}} \end{pmatrix}^2$$

We can observe a rotation matrix for a fixed angle.

$$\begin{pmatrix} \sqrt{\frac{|A_0|}{N}} & -\sqrt{\frac{|A_1|}{N}} \\ \sqrt{\frac{|A_1|}{N}} & \sqrt{\frac{|A_0|}{N}} \end{pmatrix} = \begin{pmatrix} cos(\theta) & -sin(\theta) \\ sin(\theta) & cos(\theta) \end{pmatrix}$$

$$\theta = sin^{-1}(\sqrt{\frac{|A_1|}{N}})$$

By applying the square, we observe in this case that M is a rotation matrix of 2θ:

$$M = \begin{pmatrix} cos(2\theta) & -sin(2\theta) \\ sin(2\theta) & cos(2\theta) \end{pmatrix}$$

After initializing the step, this corresponds to the register of Q:

$$|u\rangle = \sqrt{\frac{|A_0|}{N}}|A_0\rangle + \sqrt{\frac{|A_1|}{N}}|A_1\rangle = cos(\theta)|A_0\rangle + sin(\theta)|A_1\rangle$$

Each time the Grover operator is applied, the Q state is rotated by an angle 2θ:

$$|u\rangle = cos(\theta)|A_0\rangle + sin(\theta)|A_1\rangle$$

$$G|u\rangle = cos(3\theta)|A_0\rangle + sin(3\theta)|A_1\rangle$$

$$G^2|u\rangle = cos(5\theta)|A_0\rangle + sin(5\theta)|A_1\rangle$$

$$\cdots$$

$$G^t|u\rangle = cos((2t+1)\theta)|A_0\rangle + sin((2t+1)\theta)|A_1\rangle$$

Geometric figure:

Main idea:
The operator $G = H^{\otimes n} Z_{OR} H^{\otimes n} Z_f$ is composed of two reflections:

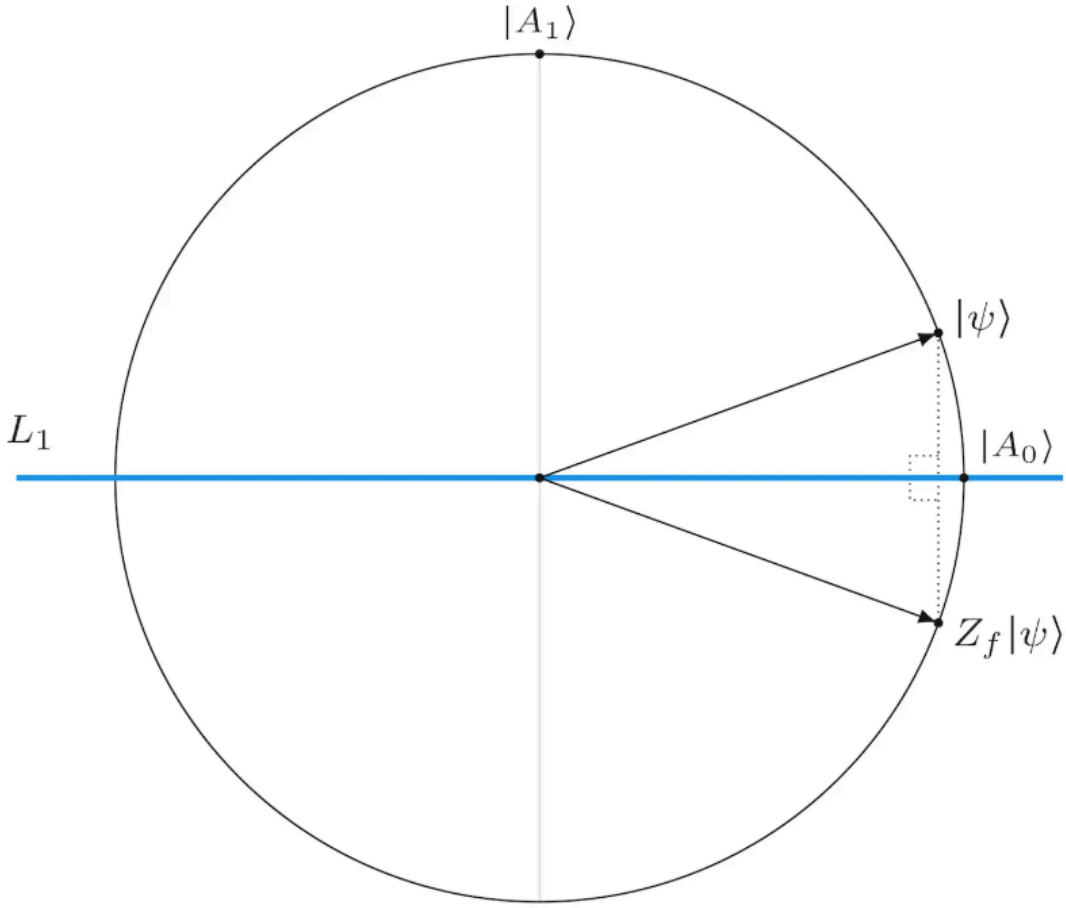
$$Z_f \text{ and } \ H^{\otimes n} Z_{OR} H^{\otimes n}$$

Composing two reflections leads to a rotation.

1. Reminder of the action of Z_f:

$$Z_f|A_0\rangle = |A_0\rangle$$

$$Z_f|A_1\rangle = -|A_1\rangle$$

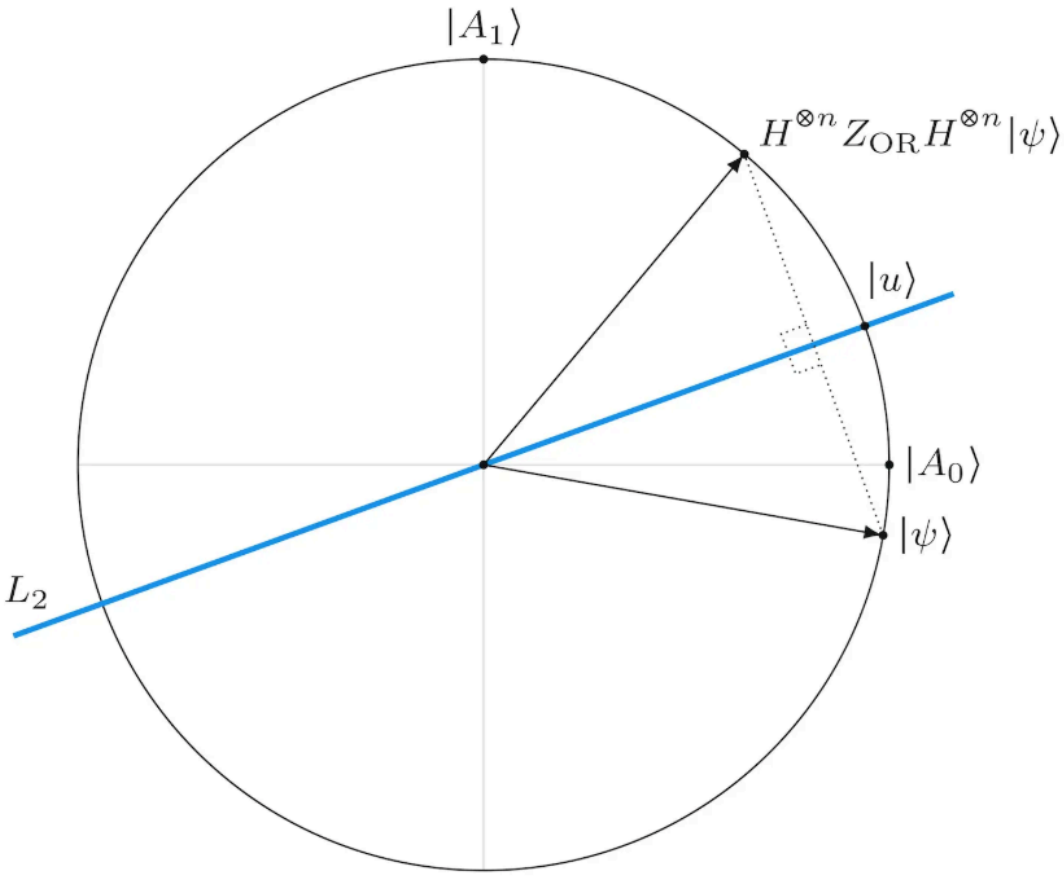


This corresponds to a reflection of $|\psi\rangle$ with respect to the line L_1 parallel to $|A_0\rangle$.

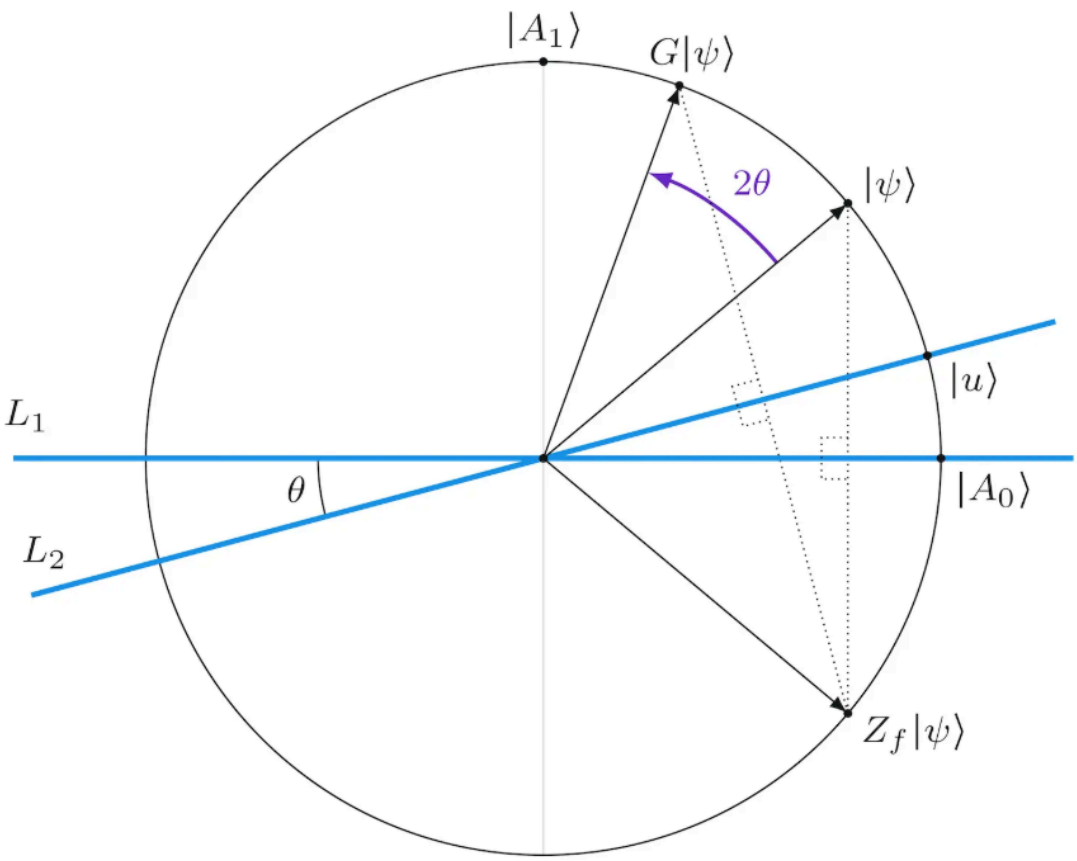
2. The operation $H^{\otimes n} Z_{OR} H^{\otimes n}$ can be expressed as follows:

$$H^{\otimes n} Z_{OR} H^{\otimes n} = 2|u\rangle\langle u| - \mathbb{I}$$

This corresponds to a reflection with respect to L_2 parallel to $|u\rangle$.



3. When two reflections are combined, the result is a rotation of twice the angle between the two lines (L_1 and L_2).



Target settings:

Consider a quantum state of the form:

$$\alpha|A_1\rangle + \beta|A_0\rangle$$

The measurements lead to a solution $x \in A_1$ with a probability $|\beta|^2$.

$$\alpha|A_1\rangle + \beta|A_0\rangle = \frac{\alpha}{\sqrt{|A_0|}} \sum_{x \in A_0} |x\rangle + \frac{\beta}{\sqrt{|A_1|}} \sum_{x \in A_1} |x\rangle$$

The output probabilities:

$$p(x) = \begin{cases} \frac{|\alpha|^2}{|A_0|} & x \in A_0 \\ \frac{|\beta|^2}{|A_1|} & x \in A_1 \end{cases} \tag{4}$$

$$P_r(A_1) = \sum_{x \in A_1} p(x) = |\beta|^2$$

What is the Q operator when Grover's algorithm is applied t times?

$$G^t|u\rangle = cos((2t+1)\theta)|A_0\rangle + sin((2t+1)\theta)|A_1\rangle,\ \theta = sin^{-1}(\sqrt{\frac{|A_1|}{N}})$$

The probability of measuring the output $x \in A_1$ after t iterations is:

$$sin^2((2t+1)\theta)$$

We want to maximize the probability, so we say that $|A_1\rangle$ is our target state.

To maximize the probability close to 1 and maximize t, we set the parameters as follows:

$$(2t+1)\theta \approx \frac{\pi}{2} \Leftrightarrow t \approx \frac{\pi}{4\theta} - \frac{1}{2} \rightarrow t = \left\lfloor \frac{\pi}{4\theta} \right\rfloor$$

Important considerations:

- t must be an integer.
- θ depends on the number of solutions.

3. Choosing the number of iterations

Unique search:

> Input: $f : \Sigma^n \rightarrow \Sigma$
> Promise: There is a unique solution $z \in \Sigma^n$ for which $f(z) = 1$, and for which $f(z) = 0$ for all other $x \neq z$.
> Output: The string z.

For a unique search, we have $s = |A_1| = 1$, and therefore:

$$\theta = sin^{-1}(\sqrt{\frac{1}{N}}) \approx \sqrt{\frac{1}{N}}$$

Substituting $\theta \approx 1/\sqrt{N}$ into the expression for t, we obtain:

$$t = \left\lfloor \frac{\pi}{4}\sqrt{N} \right\rfloor$$

We have a number of queries with complexity $O(\sqrt{N})$.

A measurement after t iterations gives a unique solution for $x \in A_1$ with a probability of:

$$p(N, 1) = sin^2((2t + 1)\theta)$$

It can be proven analytically that:

$$p(N, 1) \geq 1 - \frac{1}{N}$$

Success probabilities for Unique search

N	p(N, 1)	N	p(N, 1)
2	.5	128	.9956199
4	1.0	256	.9999470
8	.9453125	512	.9994480
16	.9613190	1024	.9994612
32	.9991823	2048	.9999968
64	.9965857	4096	.9999453

Multiple solutions:

> Input: $f : \Sigma^n \to \Sigma$
> Promise: There is a multiple solution $z \in \Sigma^n$ for which $f(z) = s$, and for which $f(z) = 0$ for all other $x \neq z$.
> Output: The string z.

$$s = 4 \to \theta = sin^{-1}(\sqrt{\frac{s}{N}})$$

For each $s \in \{1, \ldots, N\}$ the measurement probability $p(N, s)$ of finding a solution satisfies:

$$p(N, s) \geq max\left\{1 - \frac{s}{N}, \frac{s}{N}\right\}$$

Number of queries:

$$\theta = sin^{-1}(\sqrt{\frac{s}{N}}), \;\; t = \left\lfloor \frac{\pi}{4\theta} \right\rfloor$$

Each iteration of Grover's algorithm requires 1 query to evaluate f. How many

queries t depend on N and s?

$$sin^{-1} \geq x, \ \forall x \in [0,1]$$
$$\theta = sin^{-1}(\sqrt{\frac{s}{N}}) \geq \sqrt{\frac{s}{N}}$$
$$t \leq \frac{\pi}{4\theta} \leq \frac{\pi}{4}\sqrt{\frac{N}{s}}$$
$$t = O(\sqrt{\frac{N}{s}})$$

<u>What happens when we don't know the number of solutions in advance</u>?

Simple approach:

Choose a number of iterations $t \in \left\{1, \ldots, \left\lfloor \pi\sqrt{N}/4 \right\rfloor\right\}$ in a uniform random manner.

- The probability of finding a solution (if it exists) is at least 40%. (Repeat the process several times to increase the probability).
- The number of queries or evaluations of f is of complexity $O(\sqrt{N})$

More sophisticated approach:

> 1. Let $T = 1$
> 2. Run Grover's algorithm with $t \in \{1, \ldots, T\}$ chosen uniformly at random.
> 3. If a solution is found, stop and output the value of T.\
>
> Otherwise, increase T and return to step 2 (or report that there is no solution).

- The rate increases with T must be correctly balanced: a low rate requires more queries, a high rate decreases the probability of success. $T \leftarrow \left\lceil \frac{5}{4}T \right\rceil$ works.
- If the number of solutions $s \geq 1$, then the number of queries f requires $O(\sqrt{N/s})$. If there is no solution, then $O(\sqrt{N})$ queries are required.

4. Conclusion

- Grover's algorithm is asymptotically optimal
- Grover's algorithm is widely applicable
- The technique of Grover's algorithm can be generalized

5. Implementation on Qiskit

Grover Algorithm

In [1]:
```python
import math
from qiskit import QuantumCircuit
from qiskit.circuit.library import MCMTGate, ZGate, grover_operator
from qiskit.visualization import plot_distribution

from qiskit_ibm_runtime.fake_provider import FakeBrisbane
from qiskit_ibm_runtime import SamplerV2 as Sampler
```

In [2]:
```python
backend = FakeBrisbane()
```

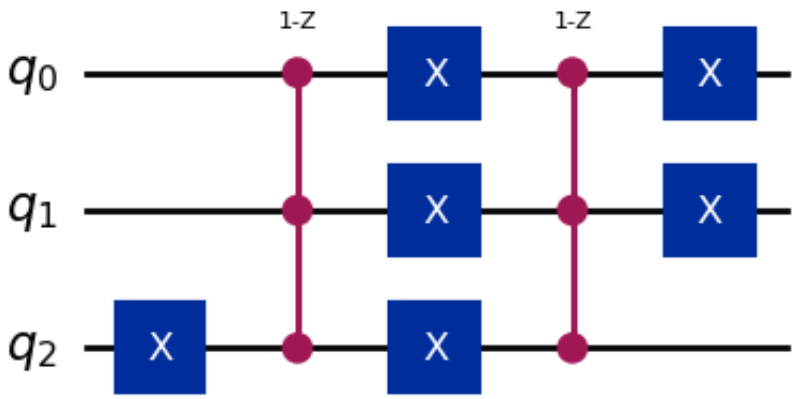
In [3]:
```python
def grover_oracle(marked_states):
    """Build a Grover oracle for multiple marked states

    Here we assume all input marked states have the same number of bits

    Parameters:
        marked_states (str or list): Marked states of oracle

    Returns:
        QuantumCircuit: Quantum circuit representing grover oracle
    """
    if not isinstance(marked_states, list):
        marked_states = [marked_states]
    #compute the number of qubits in circuit
    num_qubits = len(marked_states[0])

    qc = QuantumCircuit(num_qubits)
    # mark each target state in the input list
    for target in marked_states:
        # Flip target bit-string to match Qiski bit-ordering
        rev_Target = target[::-1]
        # Find the indices of all the '0' elements in bit string
        zero_inds = [ind for ind in range(num_qubits) if rev_Target.start
        #add a multi-controlled Z-Gate with pre- and post-applied X-Gates
        #where the target bit-string has a '0' entry
        qc.x(zero_inds)
        qc.compose(MCMTGate(ZGate(), num_qubits-1,1), inplace=True)
        qc.x(zero_inds)
    return qc
```

specific instance

Here we define the value to search. In this example, $s = 2$ the number of solution

In [4]:
```python
marked_states = ["011", "100"]

oracle = grover_oracle(marked_states)
oracle.draw("mpl")
```

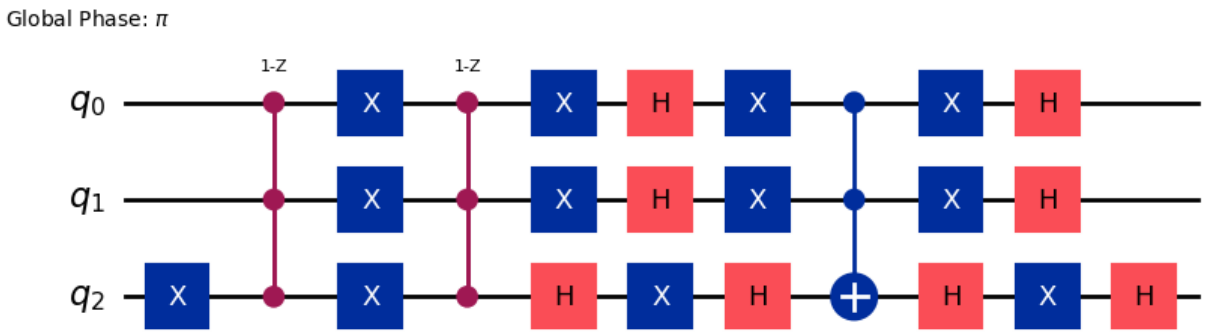
Out[4]:



GroverOperator

In [5]:
```
grover_op = grover_operator(oracle)
grover_op.draw("mpl", style = 'iqp')
```

Out[5]:



In [6]:
```
optimal_num_iterations = math.floor(math.pi/(4*math.asin(math.sqrt(len(ma
print(optimal_num_iterations)
```

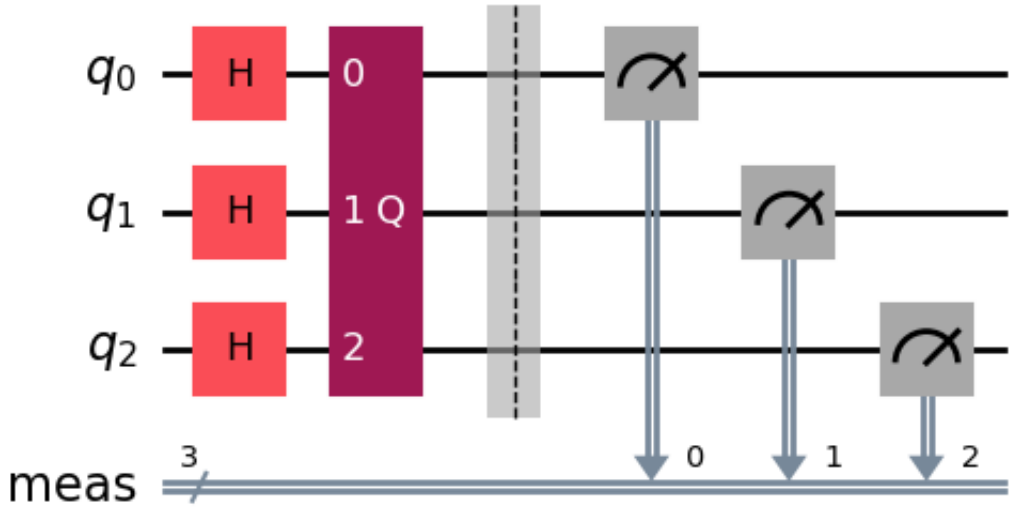
1

Full Grover circuit

Step 1: Mapping the problem

In [7]:
```
qc = QuantumCircuit(grover_op.num_qubits)
# create even superposition of all basis states
qc.h(range(grover_op.num_qubits))
#apply grover operator the optimal number of times
qc.compose(grover_op.power(optimal_num_iterations), inplace = True)
qc.measure_all()
qc.draw("mpl")
```

Out[7]:



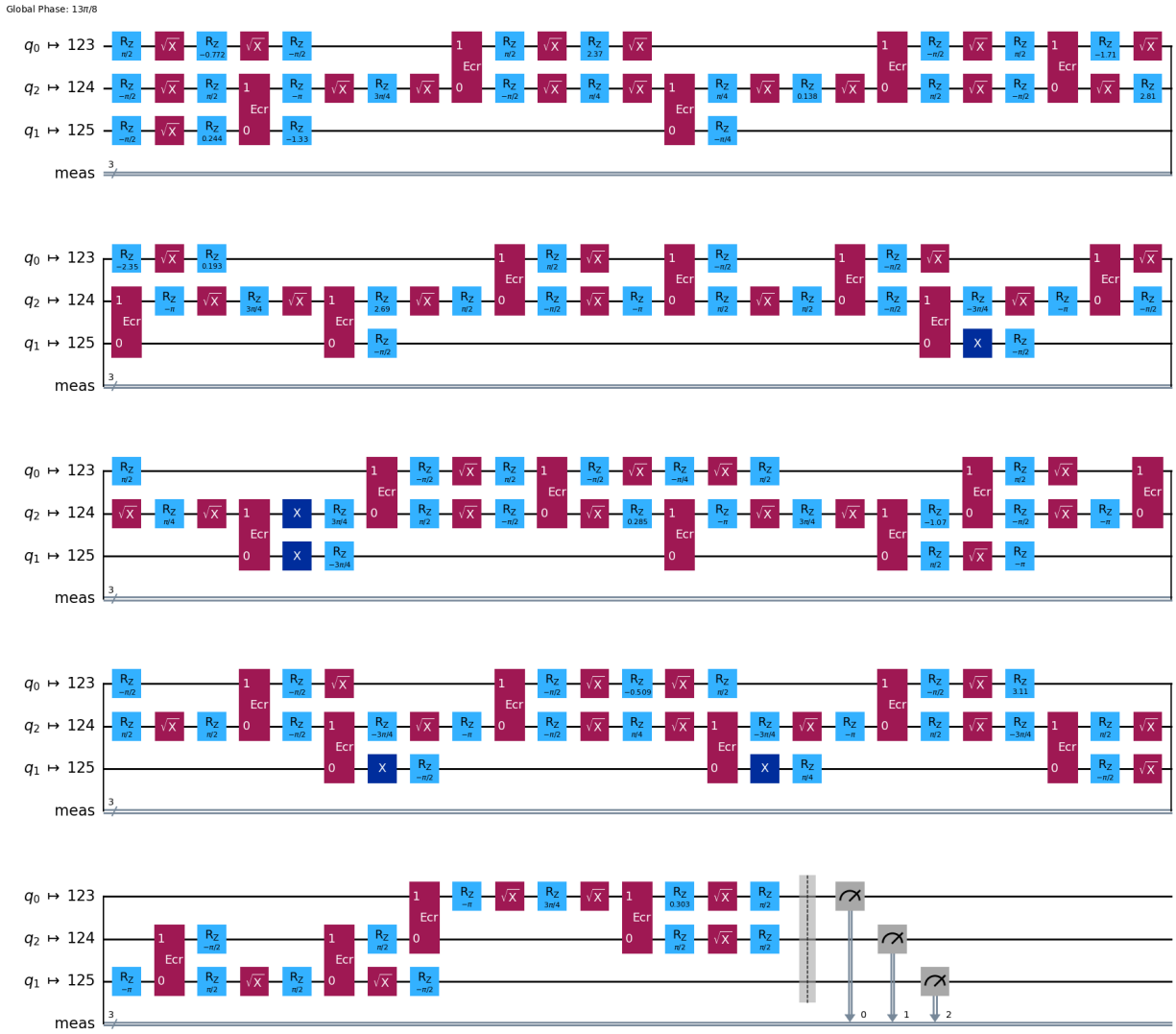
Step 2: optimize the circuit

In [8]:

```
from qiskit.transpiler.preset_passmanagers import generate_preset_pass_ma
target = backend.target
pm = generate_preset_pass_manager(target=target, optimization_level=3)

circuit_isa = pm.run(qc)
circuit_isa.draw("mpl", idle_wires =False)
```

```
(CVXPY) Aug 03 05:29:47 PM: Encountered unexpected exception importing sol
ver GLOP:
RuntimeError('Unrecognized new version of ortools (9.14.6206). Expected <
9.12.0. Please open a feature request on cvxpy to enable support for this
version.')
(CVXPY) Aug 03 05:29:47 PM: Encountered unexpected exception importing sol
ver PDLP:
RuntimeError('Unrecognized new version of ortools (9.14.6206). Expected <
9.12.0. Please open a feature request on cvxpy to enable support for this
version.')
```

Out[8]:



Step 3: execute using qiskit primitives

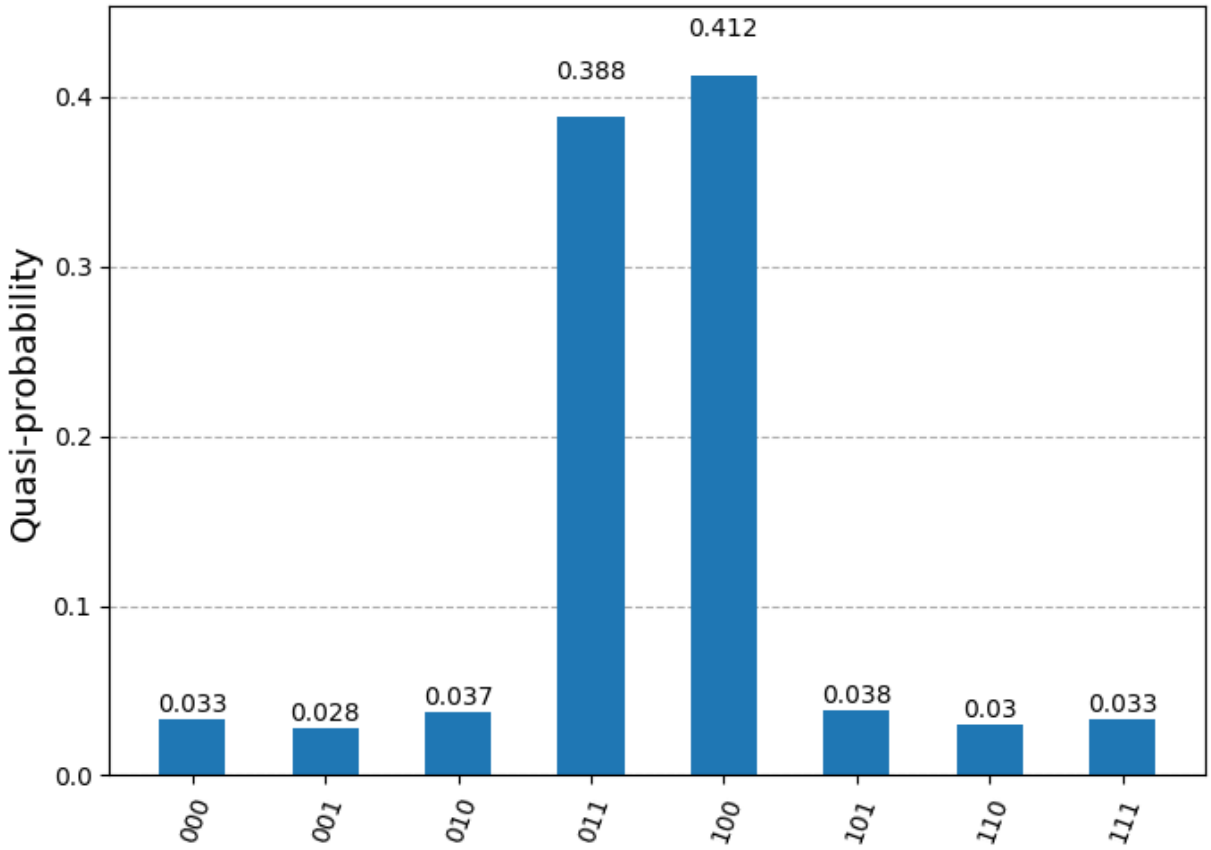
In [9]:

```
sampler = Sampler(mode=backend)
sampler.options.default_shots = 10000
result = sampler.run([circuit_isa]).result()
dist = result[0].data.meas.get_counts()
```

In [10]:

```
dist = result[0].data.meas.get_counts()
plot_distribution(dist)
```

Out[10]:



Step 4: post processing

Here we can see the two maximum probability correspond to the two searched value.

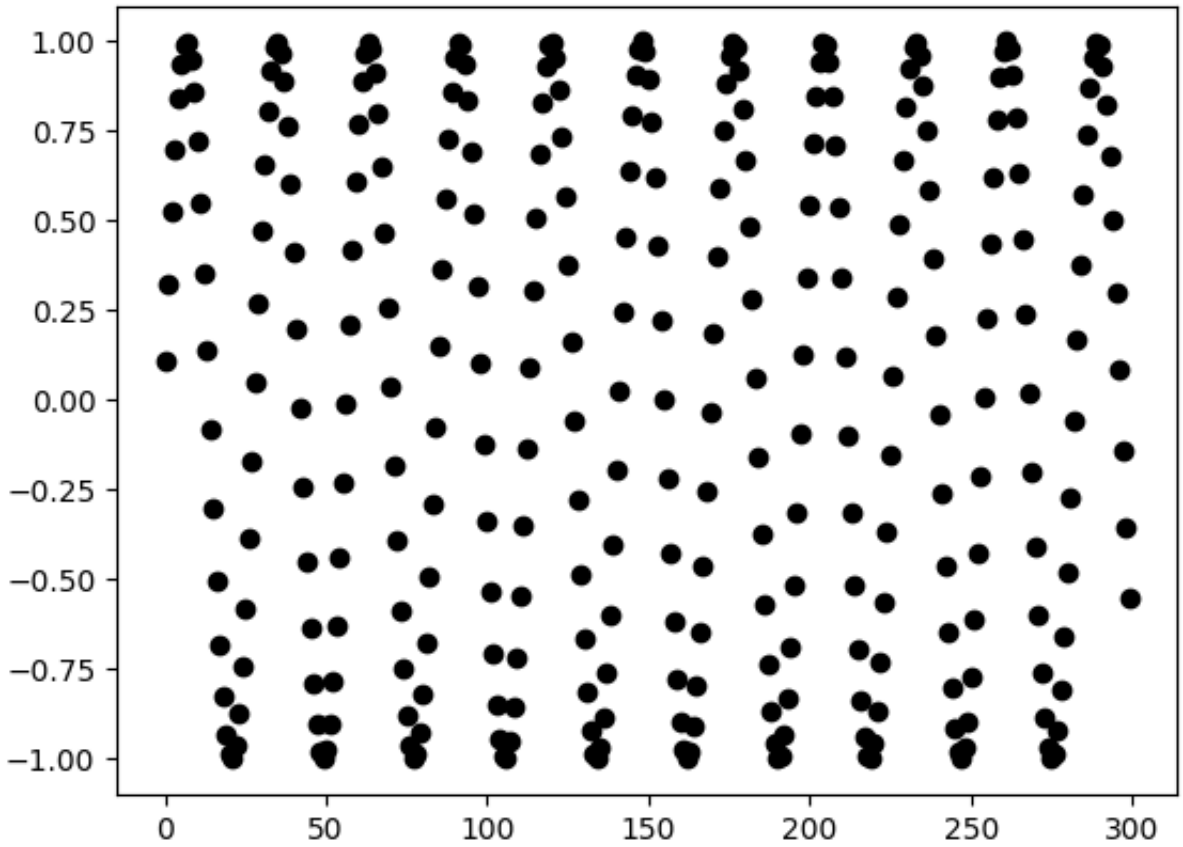
Precision of the Grover algorithm depending of the number of solutions

Single Output

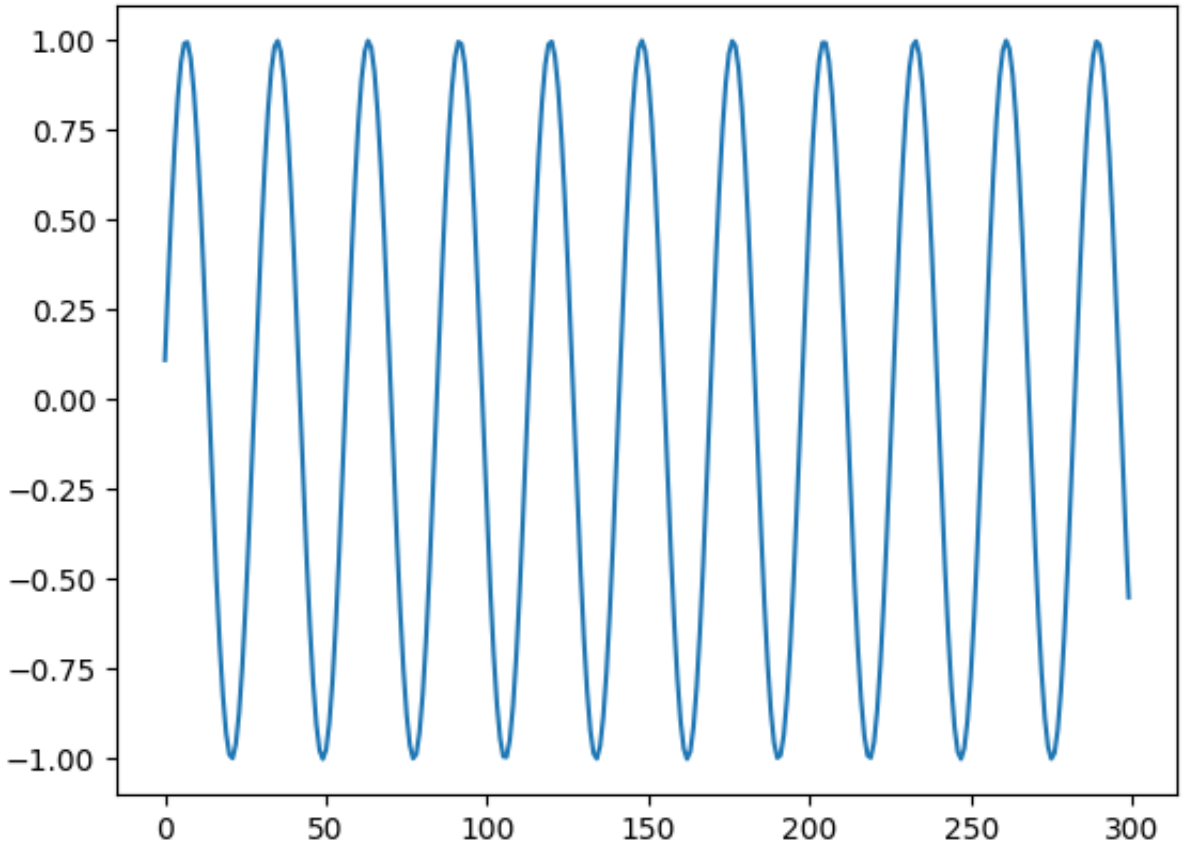
In [11]:
```
import numpy as np
import math
import matplotlib.pyplot as plt
```

In [12]:
```
theta = 1/9 # the angle theta
t = np.arange(0,300,1) # number of iterations
y = np.sin((2*t+1)*theta)

print("Scatter Plot")
plt.plot(t,y,'o', color='black')
plt.show()

print("Linear interpolation")
plt.plot(t,y)
plt.show()
```

```
Scatter Plot
```



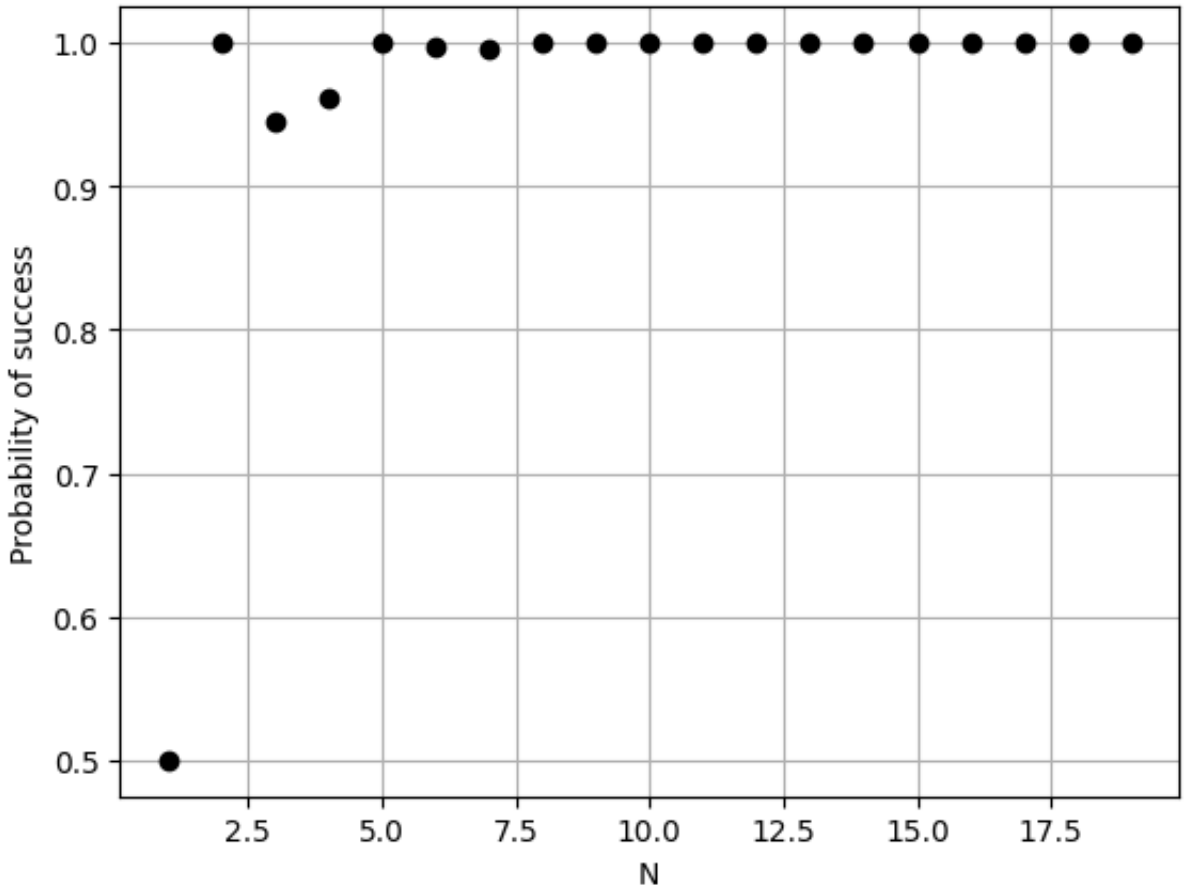

```
Linear interpolation
```



In [13]:
```
# Probability of success of Grover Algo for N in range from 0 to 2**20, f
for n in range(1,20):
    N = 2**n
    theta = np.arcsin(np.sqrt(1/N))
    t = math.floor(np.pi/(4*theta))
    p = np.sin((2*t+1)*theta)**2
```

```
    print("%d\t %12.10f" % (N, p))
    plt.plot(n,p, 'o', color='black')
plt.grid()
plt.xlabel("N")
plt.ylabel("Probability of success")
plt.show()
```

```
2        0.5000000000
4        1.0000000000
8        0.9453125000
16       0.9613189697
32       0.9991823155
64       0.9965856808
128      0.9956198657
256      0.9999470421
512      0.9994480262
1024     0.9994612447
2048     0.9999968478
4096     0.9999453461
8192     0.9999157752
16384    0.9999997811
32768    0.9999868295
65536    0.9999882596
131072   0.9999992587
262144   0.9999978382
524288   0.9999997279
```

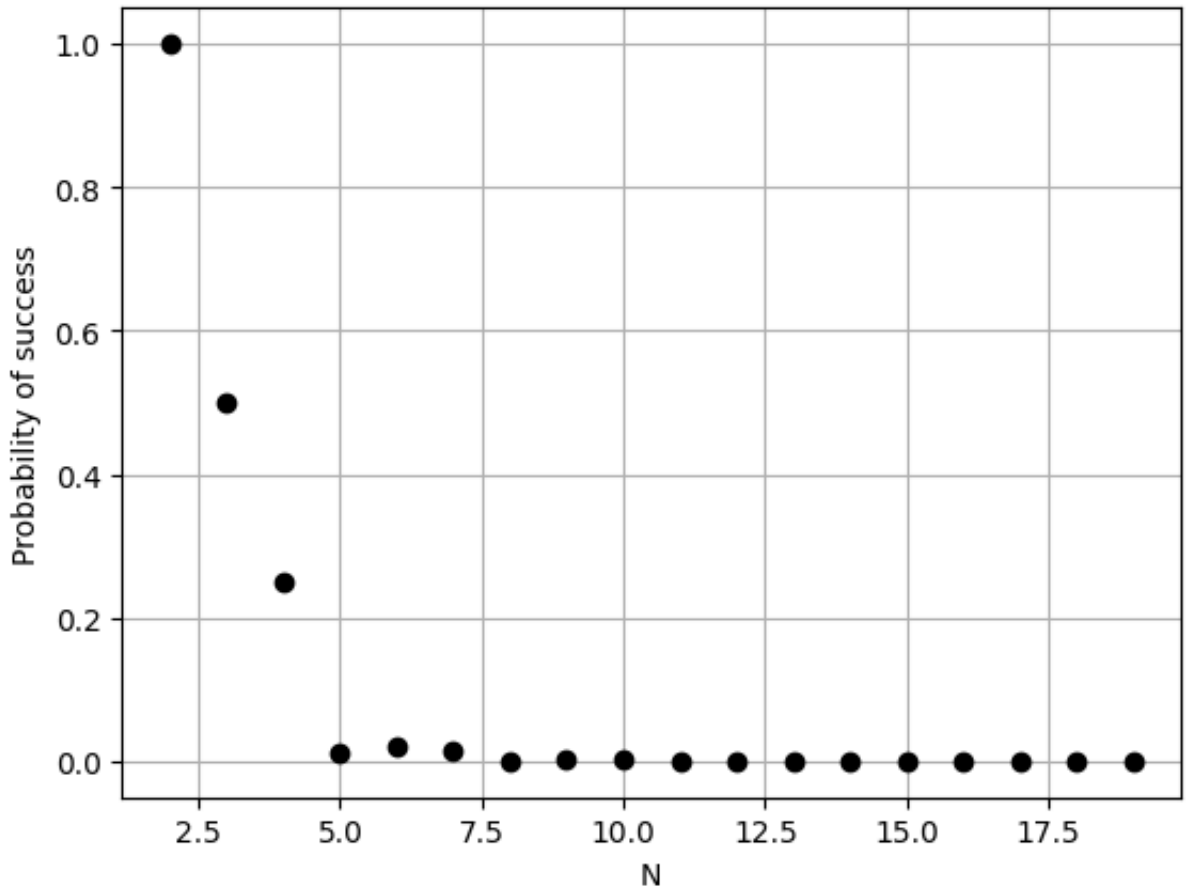


Multiple Output

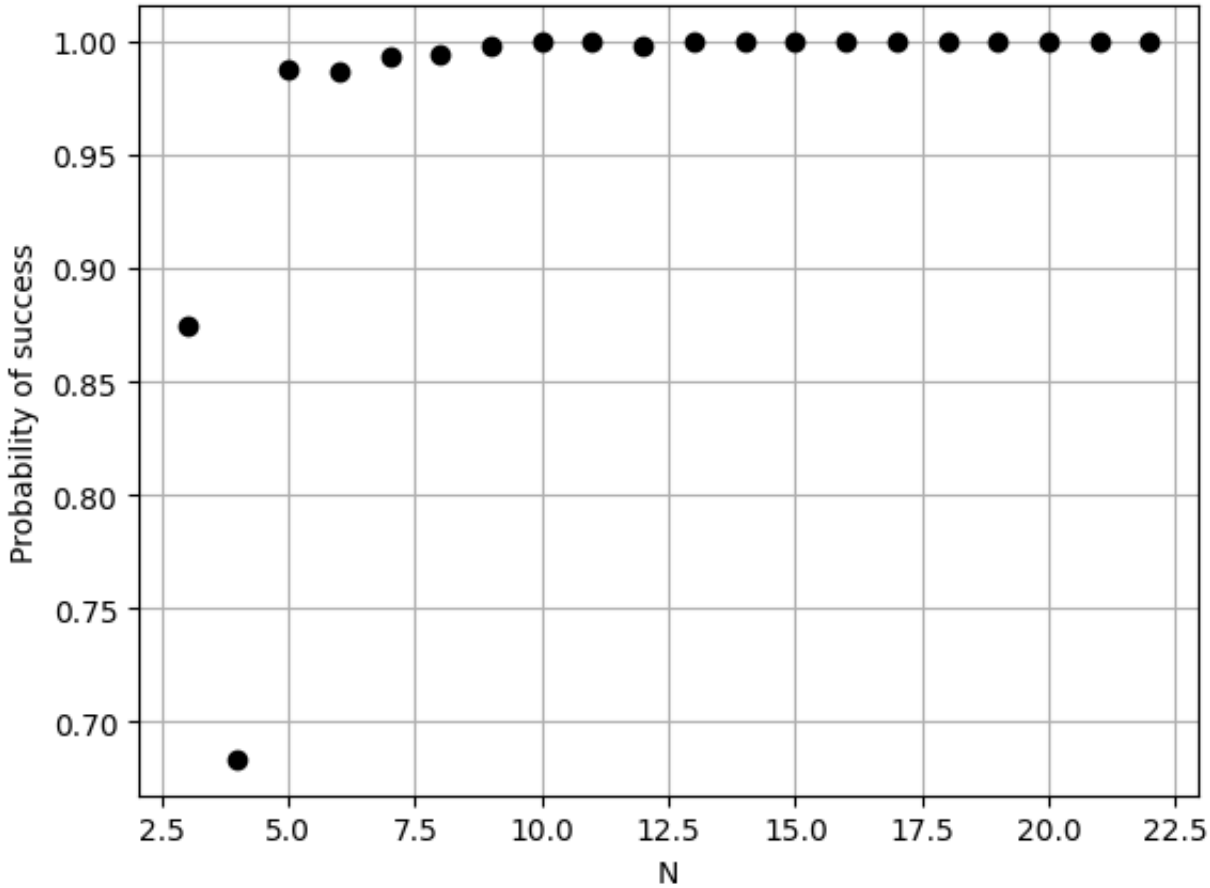
In [14]:
```
# Starting the loop with n=2 because we can't have 4 solutions
```

```python
# when n=1.

for n in range(2,20):
    N = 2**n
    wrong_theta = np.arcsin(np.sqrt((1/N)))
    correct_theta = np.arcsin(np.sqrt(4/N))
    t = math.floor(np.pi/(4*wrong_theta))
    p = np.sin((2*t+1)*correct_theta)**2

    print("%d\t %12.10f" % (N, p))
    plt.plot(n,p, 'o', color='black')
plt.grid()
plt.xlabel("N")
plt.ylabel("Probability of success")
plt.show()
```

```
4         1.0000000000
8         0.5000000000
16        0.2500000000
32        0.0122070313
64        0.0203807689
128       0.0144530758
256       0.0000705058
512       0.0019310741
1024      0.0023009083
2048      0.0000077506
4096      0.0002301502
8192      0.0003439882
16384     0.0000007053
32768     0.0000533810
65536     0.0000472907
131072    0.0000030066
262144    0.0000086824
524288    0.0000010820
```



Here we can see that the probability of sucess decrease, beacause the initilisaitlization of θ is wrong. Then when we increase the number of iterations we get the wrong probability of success.

In [15]:
```
s = 7 # number of solutions

# the loop starts with the smallest value of n that allows s solutions

for n in range(math.ceil(math.log2(s)), math.ceil(math.log2(s)+20)):
    N = 2**n
    theta = np.arcsin(np.sqrt(s/N))
    t = math.floor(np.pi/(4*theta))
    p = np.sin((2*t+1)*theta)**2
    print("%d\t %12.10f" % (N, np.sin((2 * t + 1) * theta) ** 2))
    plt.plot(n,p, 'o', color='black')
plt.grid()
plt.xlabel("N")
plt.ylabel("Probability of success")
plt.show()
```

```
8        0.8750000000
16       0.6835937500
32       0.9877929688
64       0.9869401455
128      0.9933758959
256      0.9942813445
512      0.9977678832
1024     0.9999963373
2048     0.9999257666
4096     0.9983374778
8192     0.9995465664
16384    0.9995822234
32768    0.9999531497
65536    0.9998961946
131072   0.9999998224
262144   0.9999745784
524288   0.9999894829
1048576  0.9999939313
2097152  0.9999979874
4194304  0.9999986243
```